

Automation



Challenge Title: Automated RFP Intelligence & Commercial Proposal Generation System

Challenge Reference: CSTAM-OliveSoft

Company Collaborator: OliveSoft

Assigned Mentor:

- Khalil Masmoudi - OliveSoft AI & Data Engineer

- Bilel Djemel - OliveSoft AI & Data Engineer

1. Company Introduction & Challenge Context

OliveSoft operates in the IT Services & Software Engineering sector, specializing in enterprise digital solutions, web/mobile engineering, and custom software development. To drive digital transformation and address emerging business challenges, the company continuously explores innovations in Generative AI & LLMs, Multi-Agent Orchestration (n8n), and Retrieval-Augmented Generation (RAG).

In today's rapidly evolving software market, where rapid response times and tailored commercial proposals for public/private RFPs (Appels d'Offres) are increasingly critical, we currently rely on manual market monitoring, fragmented internal archives (past projects, client history, CV databases, tech stacks), and manually formatted commercial proposals. However, as our operations scale, we are actively seeking innovative approaches to automate tender discovery, conduct AI-driven prospect research, and generate customized commercial proposal presentations.

2. Problem Statement

Our main challenge is the time-intensive, manual process of identifying relevant RFPs, gathering prospect intelligence, aligning internal expertise (CVs, past projects, tools), and drafting comprehensive commercial proposals.

When a new tender or RFP is published or submitted, our teams experience delays in evaluation, high manual research overhead, and risk of suboptimal positioning due to fragmented access to internal knowledge assets. We require a robust solution capable of automatically detecting market leads, conducting agentic prospect research, retrieving matching OliveSoft capabilities via RAG, and generating customized commercial presentations and proposals without disrupting existing workflows.

3. Technical Mission

Challenge Objective:

To design, prototype, and demonstrate an end-to-end AI-powered system that monitors market tenders, enriches context through autonomous prospect research, leverages OliveSoft's internal knowledge base via RAG, and automatically generates tailored commercial proposal presentations using n8n orchestration and Generative AI.

Participating teams must build a system evaluated across the following core phases:

Phase 1: Core Functionalities MVP (Mandatory)

Total score: 50 points

Required Features:

- **Lead & Tender Detection Engine [15 Points]:** Automated ingestion and detection of published tenders/RFPs from public datasets, simulated feeds, or structured inputs.
- **Robust RAG Knowledge Retrieval [15 Points]:** A vector-search pipeline capable of indexing and extracting knowledge from Olivesoft's internal documentation (past project references, client portfolios, CV bank, tools, and technical stacks).
- **Agentic Prospect Research [10 Points]:** Autonomous AI agent workflow to research prospect details (sector, estimated revenue, past projects, key partners, specific domain requirements).
- **Workflow Orchestration with n8n [10 Points]:** Seamless orchestration using n8n to connect detection, research, RAG retrieval, and initial document generation.

Required Deliverables :

- System architecture diagram and data pipeline flow documentation
- Source code repository (including n8n workflows and RAG implementation)
- A short video demonstration of the MVP

Phase 2: Integration & Testing

Total score: 30 Points

Required Features:

- **Generative Presentation & Proposal Output [15 Points]:** Automated drafting of commercial proposal presentations matching Olivesoft's commercial templates and branding guidelines.
- **RAG Benchmark & Retrieval Precision [10 Points]:** Benchmarking and optimizing semantic search accuracy to ensure correct CV and project matching based on tender requirements.

- **Edge Case & Data Validation Handling [5 Points]:** Graceful handling of incomplete tender specifications, noisy web research data, or missing internal references.

Required Deliverables :

- RAG benchmark, evaluation report, and test suite execution results.
- Detailed integration & API workflow documentation.

Phase 3: UI/UX & Documentation

Total score: 10 Points

Refine the project for final evaluation by improving usability and preparing all required deliverables.

Required Features:

- **Quality of Interface / Usability [10 Points]:** Intuitive web interface/dashboard allowing sales managers to view detected leads, inspect matched internal assets, review prospect research, edit generated parameters, and download proposal decks.

Required Deliverables :

- Finalized UI/UX design and user experience dashboard.
- Technical report documenting project architecture, LLM/RAG strategies, and results.
- A demonstration video showcasing full end-to-end platform features and functionality.

Note:

- **Only the selected teams will advance to the Final Pitching phase**
- **The detailed selection criteria are provided in the General CSTAM 3.0 Challenge Guide**

Phase 4: Pitching

Total score: 10 Points

Present the completed solution to the company representatives and the evaluation committee during the CSTAM 3.0 event.

Required Deliverables :

- A professional presentation supporting the pitch.
- A live demonstration of the platform.
- Q&A session with the jury members.

Optional Advanced Features

Bonus Points

- **Multi-Format Deck Exporting [+2 Points]:** Direct generation and export to editable .pptx slides and formatted .pdf documents.
- **Automated Requirements Matrix & Score [+2 Points]:** Auto-generating a matrix scoring OliveSoft's technical coverage against RFP requirements.
- **Competitor & Budget Insights Agent [+2 Points]:** Agentic estimation of prospective budget ranges and competitive positioning analysis.
- **Human-in-the-Loop AI Refinement [+2 Points]:** An interactive UI feature allowing sales teams to prompt-edit specific slide sections in real-time.
- **Multi-Source Crawler Integration [+2 Points]:** Ingestion pipelines for real-time crawling across multiple public procurement portals.

4. Recommended Skills

You're encouraged to bring your own ideas, but familiarity with the following helps:

- **Hardware / Infrastructure:** Cloud
- Services, Docker Containers, Serverless Deployments.
- **Software & Frameworks:** n8n Workflow Automation, Python, LangChain / LlamaIndex, Fast API, React / Next.js / Tailwind CSS, Vector Databases (Qdrant, ChromaDB, Pinecone, or pgvector).
- **Protocols & Models:** REST APIs, Webhooks, OpenAI / Anthropic APIs, Open-Source LLMs (Ollama / vLLM), JSON-Schema formatting.

5. Rewards & Opportunities

- **Number of Winners:** 3 teams
- **Awards**
 - **First Place Team:** Summer internships and PFE 2027 opportunities at OliveSoft
 - **Second Place Team:** Certificate of Recognition
 - **Third Place Team:** Certificate of Recognition

Note:

- **Additional rewards may be announced later**
- **Teams will also compete for the Global CSTAM Book Grand Prizes Top 3 across all challenges**